Template Document

Academic Support Agent

February 19, 2025

Abstract

[flattitle] v
documentclass `\shtsetup`

[flattitle] We elaborate on the usage of document class v, including its compiling process, class options and provided functionalities. We hope this document can help the reader to quickly typeset his or her thesis.

is intended to cover all format requirements of ShanghaiTech graduate/undergraduate thesis with minimal function set and introduce no restriction to the user. Specifying documentclass to is su icient for the basic typesetting requirements. This document class uses `\shtsetup` to uniformly set up mandatory information and only provides minimum auxiliary commands to meet all format requirements. Users can import other packages and functionalities according to their writing habits.

[ ]

$R$the gas constant

$C_v$specific heat capacity at constant volume

$C_p$specific heat capacity at constant pressure

Especific total energy

$k$thermal conductivity

S_{ij}deviatoric stress tensor

τ_{ij}viscous stress tensor

δ_{ij}Kronecker tensor

[ ]

CFDComputational Fluid Dynamics

CFLCourant-Friedrichs-Lewy

WENOWeighted Essentially Non-oscillatory

ZNDZel'dovich-von Neumann-Doering

[&]

Δdifference

∇gradient operator

$\delta^{\pm}$upwind-biased interpolation scheme

(**S**hang**h**ai**T**ech University **THESIS**) LaTeXv
mskbxkmfxqpt.tex
shtthesis.cls DocStrip LaTeX3 issues pull requests `https://github.com/lirundong/shtthesis`

serif *italic* sans serif hyperref `typewriter`
[<bar>]<baz> `[]` `{}` `<>` `<>` `\foo` `bar=this-arg` `baz=` [this-arg]

1

CTAN[1] TEX Live TEX Live `tlmgr` '' tlmgr install shtthesis CTAN '' tlmgr update shtthesis
CTAN shanghaitech-emblem.pdf[2] thesis.tex reference.bib bib mskbxkmfxqpt.tex

thesis.tex
reference.bib
shanghaitech-emblem.pdf

2

XeLaTeX LuaLaTeX pdfLaTeX TeX Live `latexmk` Windows Linux TeX Live (`https://www.tug.org/texlive/`) macOS MacTeX (`https://www.tug.org/mactex/`) CTEXunamused overleaf 1 CTEXoverleaf issues

`latexmk` Windows CMD XeLaTeX '' latexmk -pdfxe LuaLaTeX '' latexmk -pdflua

XeLaTeX LuaLaTeX macOS XeLaTeX PDF Windows LuaLaTeX LuaHBTeX Emoji hot-pepperpearkeycap-tensyringesweat-dropscow-facebeer-mug[3] LuaHBTeX TeX Live 2020 LuaLaTeX TeX Live 2019 LuaHBTeX '' latexmk -pdflua -pdflualatex=lualatex-dev

3

thesis.tex [< >]shtthesis `{bachelor | master | doctor}` `anonymous` `print` cexbook ,

3.1

v bachelor master doctor

bachelor2019 Word LATEX v Word
2020

Latex word Word

`comfort` [bachelor, comfort]shtthesis

3.2 `anonymous`

`anonymous` [anonymous]shtthesis *** `\shtsetup` `anonymous-str` anonymous-str = XXX, `\shtsetup`
4

3.3 `print`

`print` URL 0.63 [print]shtthesis

3.4 ctexbook

ctexbook ctexbook `fontset` fontset Windows `fontset=windows` macOS `fontset=mac` [fontset=windows]shtthesis
1 Linux/UNIX / Source Han Noto CJK Fandol / / -GBK Fandol /

Fandol TEX Live 9[4] [5] -GBK[6] -GBK[7] (GBK) Linux/UNIX

[1] `https://www.ctan.org/pkg/shtthesis`
[2] `https://github.com/lirundong/shtthesis/raw/v0.3/shanghaitech-emblem.pdf`
[3] LuaHBTeX .....
[4] `https://source.typekit.com/source-han-serif/cn/`
[5] `https://github.com/adobe-fonts/source-han-sans`
[6] `https://www.foundertype.com/index.php/FontInfo/index/id/137`
[7] `https://www.foundertype.com/index.php/FontInfo/index/id/128`

Table 1:

(a)

Windows	macOS	Linux/UNIX
		→ Fandol
		→ Fandol
		-GBK → Fandol
		-GBK → Fandol

(b)

Serif	Sans Serif	Typewriter
TEX Gyre Termes	TEX Gyre Heros	TEX Gyre Cursor

4

/ bachelor2019 \shtsetup \documentclass \begin{document} key=value \shtsetup
degree-name = , degree-name* = Master of Science in Engineering, secret-level = , title = v , title* = A User's Guide
to
v, keywords = LATEX, keywords* = ShanghaiTech University, Thesis, LATEX, author = , author* = Li Rundong, institution = , institution* = School of Information Science and Technology
ShanghaiTech University, supervisor = , supervisor* = Professor Fan Rui, supervisor-institution = , discipline-level-1 = , discipline-level-1* = Computer Science and Technology, date = 2020 6 , date* = June, 2020, bib-resource = reference.bib,
title = v
, title* = A User's Guide to
v, keywords = LATEX, keywords* = ShanghaiTech University, Thesis, LATEX, date = 2020 06 , date* = 06 / 2020, author = , author* = Rundong Li, author-id = 36273800, entrance-year = 2017, institution = , institution* = School of Information Science and Technology, supervisor = , supervisor* = Rui Fan, discipline = , discipline* = Computer Science and Technology, bib-resource = reference.bib,
\shtsetup key=value key – degree-name * key degree-name* value \shtsetup

4.1

degree-name degree-name* 2

Table 2: shtsetup

key	value	
degree-name		
degree-name*	Doctor of Philosophy	*Doctor of Philosophy Master of Natural Science Master of Science in Engineering*

4.2

title title* secret-level XXX

4.3

author author* / institution institution* discipline-level-1 discipline-level-1* discipline discipline*
author = , author* = Li Rundong,
author = , author* = Rundong Li,

Figure 1

institution institution* institution = , institution* = School of Information Science and Technology ShanghaiTech University,
institution = , institution* = School of Information Science and Technology,

4.4

supervisor supervisor* supervisor-institution supervisor supervisor* Professor supervisor-institution supervisor = , supervisor* = Professor Fan Rui, supervisor-institution = ,

4.5

date date* date = 2020 6 , date* = June, 2020,
date = 2020 06 , date* = 06 / 2020,

5

`anonymous`

5.1

Abstract

...

English abstract content...
abstract abstract* `\shtsetup` `flattitle` title title*

Abstract

[flattitle] ...

`\shtsetup` keywords keywords* v value keywords keywords* , keywords = LaTeX, keywords* = ShanghaiTech University, Thesis, LaTeX,

5.2

`\makeindices` 3 subsection PDF `\caption`
nomenclatures [< >] [< >]< >< >

< >< > nomenclatures `\item < >` section `\head` `\item` `\item < >` nomenclatures mskbxkmfxqpt.tex v `\item < >`

6

LaTeX GB/T 7714-2015 -

6.1

mathtools . . .

$$P(A|B) = \frac{P(A)P(B|A)}{P(B)} \tag{1}$$

`\eqref` amsmath (1)

unicode-math LaTeX Unicode `$\oiint$` Unicode `$$` `U+0222F` clerkma2013unicode Unicode

$$x[(FxGx)Hx] \tag{2}$$

$$yx[xyxx] \tag{3}$$

$$2005/06/28ver :$$

$$2005/06/28ver :$$

$$2005/06/28ver :$$

$$(PQ)(PQ) \tag{7}$$

6.2

amsthm 3 tex

Table 3:

theorem	lemma	corollary	proposition	conjecture
definition	axiom	example	problem	exercise
remark				

[Unicode-CJK] Unicode
Unicode
Unicode
Unicode (codepoints) `U+4F60` `U+597D`
Unicode (codepoints) `U+3053` `U+3093` `U+306B` `U+3061` `U+306F`
Unicode (`https://home.unicode.org/`)

6.3

biblatex biblatex-gb7714-2015 GB/T 7714-2015 -

`\cite` [**?**, **?**] `\inlinecite` Bohan1928

- `\citet` Bohan1928 `\citep` yuan2012lana `\citeauthor` niu2013zonghe `\citeyear` walls2013drought `\shtsetup` bib-resource reference.bib bib-resource = reference.bib,
key bravo1990comparative chen1980zhongguo bib @incollectionchen1980zhongguo, author = and and and and , key = Chen2 Jing Ao Zhang1 Hui Ming Zhu1 Shi Xing Zhao4 Zhen Wang2 Zhen Gang, title = , editor = , booktitle = , address = , publisher = , year = 1980, pages = 56–114,

6.4

enumerate itemize

1.
2.
3.
4.

`\begin{enumerate}` short label +

1.
2.

3.

4.

`resume*`

5.

6.

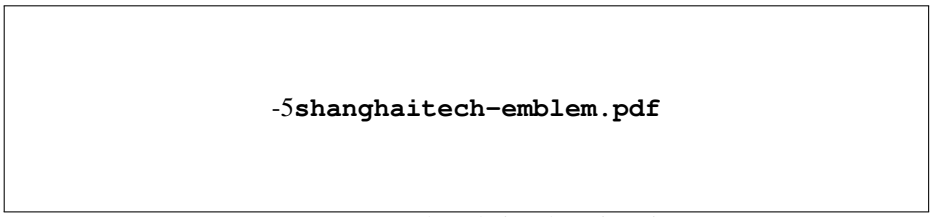
7.

8.

description

6.5

bicaption **??**



School logo of ShanghaiTech University

bicaption 2020 bicaption `list=off` [list=off]bicaption

7

`\makebiblio` bib 6.3

...

acknowledgement resume publications patents projects publications* patents* projects* `anonymous` 3.2

7.1

acknowledgement .....

7.2

resume publications publications* patents patents* projects projects* .....

.....

.....

ucasthesis B[8] [9]

[8] `https://github.com/mohuangrui/ucasthesis/blob/master/Tex/Appendix.tex`

[9] `http://xh.5156edu.com/page/z4745m2559j18770.html`

8

$$\begin{cases} \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho V) = 0 \\ \frac{\partial (\rho V)}{\partial t} + \nabla \cdot (\rho V V) = \nabla \cdot \sigma \\ \frac{\partial (\rho E)}{\partial t} + \nabla \cdot (\rho E V) = \nabla \cdot (k \nabla T) + \nabla \cdot (\sigma \cdot V) \end{cases} \tag{8}$$

$$\frac{\partial}{\partial t} \int\limits_{\Omega} u \, d\Omega + \int\limits_{S} n \cdot (uV) \, dS = \dot{\phi} \tag{9}$$

$$L\{f\}(s) = \int_{0^-}^{\infty} f(t) e^{-st} \, dt, \; L\{f\}(s) = \int_{0^-}^{\infty} f(t) e^{-st} \, dt$$

$$F\big(f(x + x_0)\big) = F\big(f(x)\big) e^{2\pi i \xi x_0}, \; F\big(f(x + x_0)\big) = F\big(f(x)\big) e^{2\pi i \xi x_0}$$

Ordinary math	$A, F, L, 2, 3, 5, \sigma$	`\symup`	$A, F, L, 2, 3, 5, \sigma$
`\symbf`	$A, F, L, 2, 3, 5, \sigma$	`\symit`	$A, F, L, 2, 3, 5, \sigma$
`\symsf`	$A, F, L, 2, 3, 5, \sigma$	`\symtt`	$A, F, L, 2, 3, 5, \sigma$
`\symfrak`	$A, F, L, 2, 3, 5, \sigma$	`\symbb`	$A, F, L, 2, 3, 5, \sigma$
`\symcal`	$A, F, L, 2, 3, 5, \sigma$	`\symscr`	$A, F, L, 2, 3, 5, \sigma$

9

ucasthesis[10] v0.2.0 thuthesis[11] `\thusetup` v0.3.0 fduthesis[12]

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[10] `https://github.com/mohuangrui/ucasthesis` GPLv3
[11] `https://www.ctan.org/pkg/thuthesis` LPPL-1.3c
[12] `https://www.ctan.org/pkg/fduthesis` LPPL-1.3c